Azulene is an organic natural product comprised of fused seven- and five-membered rings containing a conjugated pi system. The aromaticity of azulene's 10π electron system is most evident when examining its resonance form, which is represented by fused aromatic tropylium and cyclopentadiene ions (Figure 1). This electronic feature is primarily responsible for azulene's abnormally large dipole moment for a hydrocarbon (1.08 D), differential chemical reactivity of its two rings, and striking, deep blue color. As a naphthalene isostere, azulene-containing compounds have been identified with anti-inflammatory and anti-infectious (bacteria, fungal, viral, cancer) properties. The unique molecular and electronic scaffold of azulene continues to make it an interesting target within the drug development and optoelectronic communities.

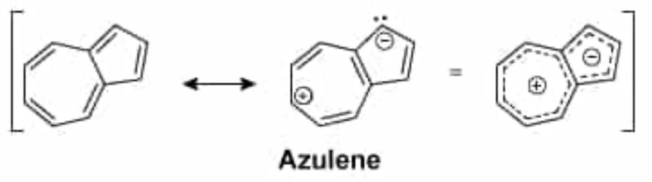


Figure 1 Azulene resonance form

The azulene-related compounds α-guaiene and guaiazulene (Figure 2) are natural products that are biosynthetically derived from the same 15-carbon metabolite, farnesyl pyrophosphate. These two materials are prepared through divergent biosynthetic pathways and have quite different properties. α-Guaiene is one of several nonaromatic, sesquiterpene isomers isolated from guaiac wood oil that have found use within the flavoring and fragrance industries. In contrast, guaiazulene is an alkylated azulene variant that shares a common aromatic core and electronic properties. Guaiazulene-derived compounds have been developed in recent years as potential treatments for a wide range of diseases and conditions.

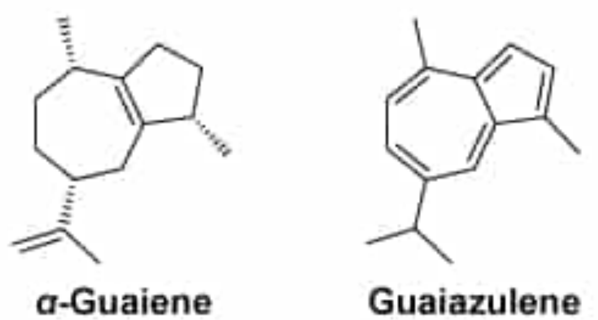


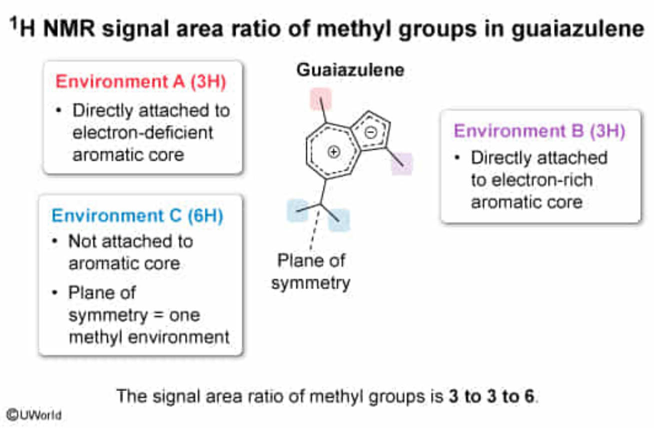
Figure 2 Azulene-related natural products

In the ^{1}H NMR spectrum of guaiazulene, what is the signal area ratio for proton environments for methyl carbons?

- A. 3 to 3 to 3 to 3 [15%]
- B. 6 to 6 [7%]
- C. 3 to 3 to 6 [64%]
- D. The ^{1}H NMR spectrum of guaiazulene has only one signal corresponding to methyl groups. [12%]

Correct
64% answered correctly

Explanation:



In proton (^{1}H) nuclear magnetic resonance (NMR) spectroscopy, samples are placed in an external magnetic field B_0 and radio waves induce transitions between low- and high-energy nuclear spin states. The signals generated indicate the:

- Number of chemical environments
- Chemical shift
- Ratio of signal intensities (integration)
- Spin-spin splitting of signals (multiplicity)

The chemical environment of a proton is influenced by proximal atoms (ie, nearby electronegative atoms or functional groups). **Symmetrical molecules** may have **fewer unique chemical environments** since protons on different carbon atoms can have identical proximity to influencing functional groups.

In a ^{1}H NMR spectrum, the area of a signal (integration) is proportional to the number of protons in the molecule that generate the signal. Therefore, the **integration** ratio of signals is **proportional to** the **ratio of protons** in the **respective chemical environments**.

Guaiazulene has four methyl groups: Two are bonded directly to the aromatic azulene core, one is on the seven-membered ring, and one is on the five-membered ring. Given the differences in the electronic environment between the two rings, the ring methyl groups are nonequivalent and appear as **two unique 3H signals**.

The other two methyl groups are in the isopropyl group, which contains a plane of symmetry and can freely rotate about the sigma bond connecting it to the azulene core. Isopropyl methyl groups are magnetically equivalent and will contribute to **one 6H signal**.

Therefore, the signal area ratio for methyl groups in guaiazulene is **3 to 3 to 6**.

(Choice A) A signal area ratio of 3 to 3 to 3 to 3 indicates four methyl groups, each in a *unique* magnetic environment.

(Choice B) A signal area ratio of 6 to 6 indicates two proton environments, each with a pair of magnetically equivalent methyl groups (2 × 3H = 6H). Because methyl groups bonded to the azulene core are attached to electronically different rings and at positions that do not confer a plane of symmetry, ring methyl groups are *inequivalent* and appear as unique signals.

(Choice D) One methyl group signal is only possible if all methyl groups are in a single magnetic environment. The methyl groups in guaiazulene are *not* in a single environment.

Educational objective:

In proton (^{1}H) NMR spectroscopy, the integration ratio of signals is proportional to the ratio of protons in a chemical environment. Functional groups with a plane of symmetry have fewer total chemical environments and signal integration ratios that reflect the symmetry.

Time spent:0 sec

QID:404131

Subject:
Organic Chemistry

System:
Separation Techniques, Spectroscopy, and Analytical Methods

Azulene is an organic natural product comprised of fused seven- and five-membered rings containing a conjugated pi system. The aromaticity of azulene's 10π electron system is most evident when examining its resonance form, which is represented by fused aromatic tropylium and cyclopentadiene ions (Figure 1). This electronic feature is primarily responsible for azulene's abnormally large dipole moment for a hydrocarbon (1.08 D), differential chemical reactivity of its two rings, and striking, deep blue color. As a naphthalene isostere, azulene-containing compounds have been identified with anti-inflammatory and anti-infectious (bacteria, fungal, viral, cancer) properties. The unique molecular and electronic scaffold of azulene continues to make it an interesting target within the drug development and optoelectronic communities.

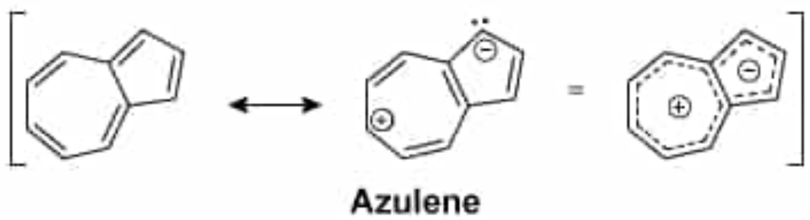


Figure 1 Azulene resonance form

The azulene-related compounds α-guaiene and guaiazulene (Figure 2) are natural products that are biosynthetically derived from the same 15-carbon metabolite, farnesyl pyrophosphate. These two materials are prepared through divergent biosynthetic pathways and have quite different properties. α-Guaiene is one of several nonaromatic, sesquiterpene isomers isolated from guaiac wood oil that have found use within the flavoring and fragrance industries.

In contrast, guaiazulene is an alkylated azulene variant that shares a common aromatic core and electronic properties. Guaiazulene-derived compounds have been developed in recent years as potential treatments for a wide range of diseases and conditions.

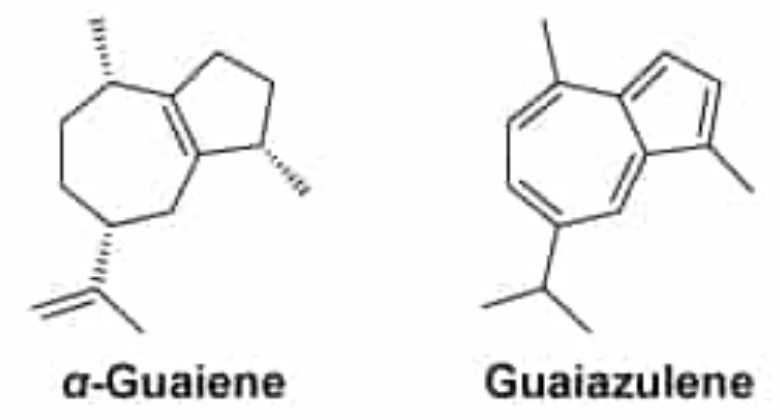
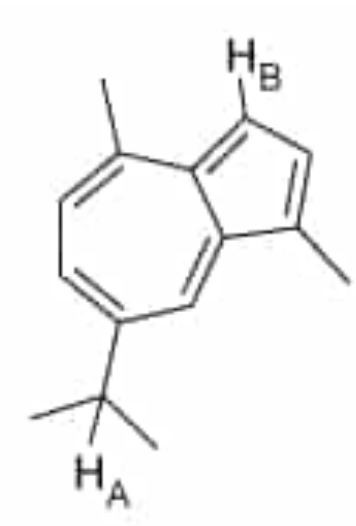


Figure 2 Azulene-related natural products

The structure of guaiazulene with annotated protons is shown below.



The chemical shift of H_B (7.22 ppm) is larger than the chemical shift of H_A (3.08 ppm) as a result of:

○ A. the inductive effect. [29%]

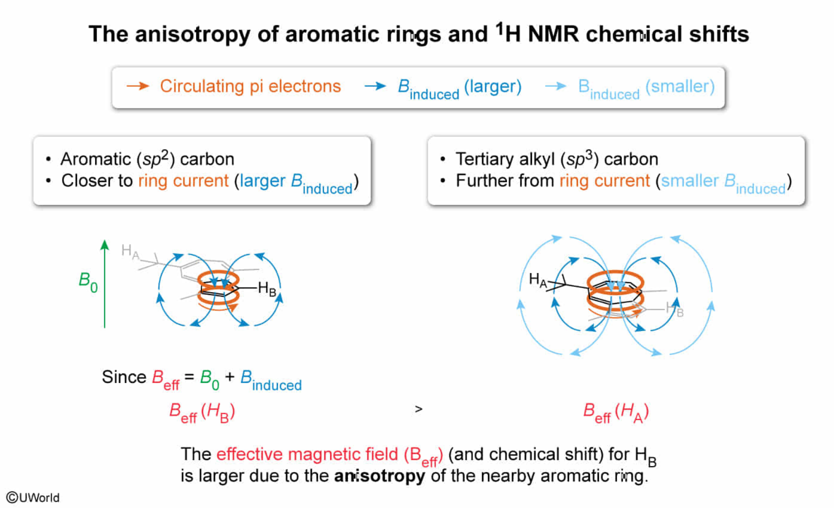
B. steric hindrance. [18%]

C. resonance. [35%]

D. the anisotropy of the aromatic ring. [16%]

Correct
16% answered correctly

Explanation:



In proton (^{1}H) nuclear magnetic resonance (NMR) spectroscopy, protons in alkyl groups are among the most shielded and normally have chemical shifts (δ) of δ 0.9–1.4. Higher-order (eg, tertiary) alkyl groups have greater chemical shifts than lower-order (eg, primary) alkyl groups.

Proton chemical shifts are also influenced by the anisotropic effect, where the **circulation of nearby pi electrons** generates an induced magnetic field $B_{induced}$ that *either* increases or decreases the effect of shielding (depending on the location of the nuclei relative to the vector of $B_{induced}$). Protons directly attached to aromatic rings are positioned in a region where $B_{induced}$ is aligned with the external magnetic field B_0, leading to a much larger effective magnetic field B_{eff} and greater chemical shift range (δ 7–8 ppm).

In the annotated structure of guaiazulene provided in this question, proton H_A is attached to the central sp^3 carbon of the isopropyl group—an alkyl environment. Being bonded to a more-substituted tertiary carbon atom, the chemical shift of H_A should be near the higher end of the alkyl range (δ 1.4–1.8 ppm). The nearby proximity of the aromatic further contributes to deshielding proton H_A. In contrast, proton H_B is bonded directly to an sp^2 carbon within an aromatic ring. Due to the **anisotropy of the aromatic ring**, the relative location of proton H_B leads to $\boldsymbol{B_{induced}}$ **reinforcing** $\boldsymbol{B_0}$, a much larger B_{eff}, and a **greater chemical shift for** $\mathbf{H_B}$.

(Choice A) The inductive effect influences the chemical shift through the donation or withdrawal of electron density in a sigma bond network. Since alkyl and aryl groups are *both* inductively donating groups, this effect does not sufficiently explain the large observed difference in chemical shift for protons H_A and H_B.

(Choice B) Steric hindrance is a principle where highly branched structural regions lead to crowding and reduced accessibility, which alone does not directly influence the chemical shift of proton environments in NMR.

(Choice C) The concept of resonance involves electron density delocalization through two or more unique Lewis structures. If resonance was primarily influencing the chemical shifts of protons H_A and H_B, the chemical shift of H_A would be predicted to be *larger* than H_B, as H_A is near the electron-deficient (cationic) seven-membered ring and H_B is near the electron-rich (anionic) five-membered ring.

Educational objective:
The chemical shift of a proton nuclear magnetic resonance (NMR) environment is affected by many factors, including the anisotropy of induced magnetic fields generated by the movement of pi electrons. Using the structure of a molecule, relative chemical shift relationships can be identified.

Time spent:0 sec

Subject:
Organic Chemistry

QID:404132

System:
Separation Techniques, Spectroscopy, and Analytical Methods

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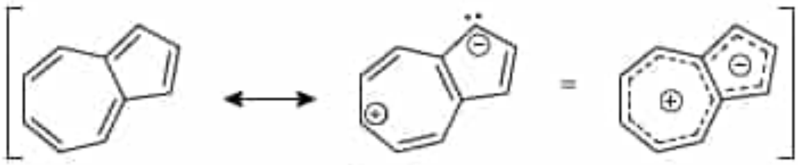


Azulene

Figure 1 Azulene resonance form

The azulene-related compounds α-guaiene and guaiazulene (Figure 2) are natural products that are biosynthetically derived from the same 15-carbon metabolite, farnesyl pyrophosphate. These two materials are prepared through divergent biosynthetic pathways and have quite different properties. α-Guaiene is one of several nonaromatic, sesquiterpene isomers isolated from guaiac wood oil that have found use within the flavoring and fragrance industries. In contrast, guaiazulene is an alkylated azulene variant that shares a common aromatic core and electronic properties. Guaiazulene-derived compounds have been developed in recent years as potential treatments for a wide range of diseases and conditions.

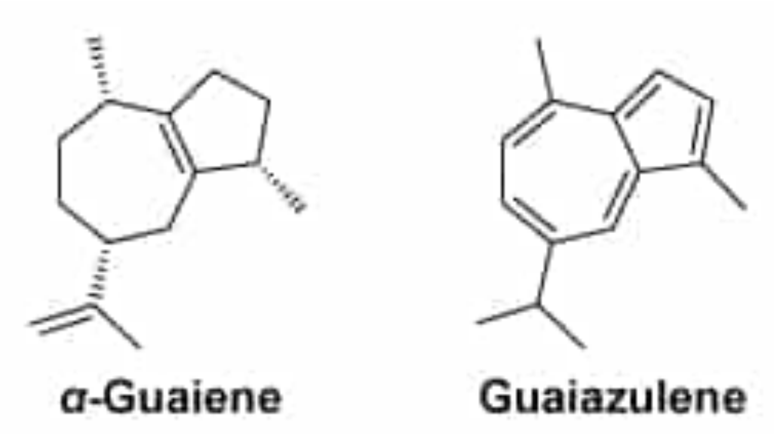


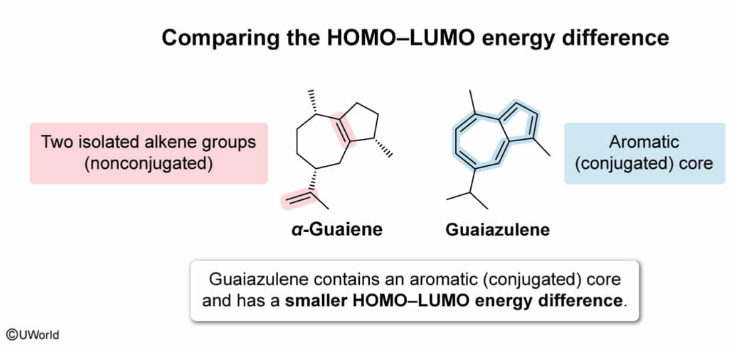
Figure 2 Azulene-related natural products

Based on the passage, what can be inferred about the relative HOMO-LUMO energy difference for α-guaiene and guaiazulene?

- ✔ ◉ A. The HOMO-LUMO energy difference is smaller for guaiazulene [43%]
- ○ B. The HOMO-LUMO energy difference is larger for guaiazulene [43%]
- ○ C. The HOMO-LUMO energy difference is the same for α-guaiene and guaiazulene [6%]
- ○ D. Nothing can be inferred about the relative HOMO-LUMO energy difference for these compounds [6%]

Correct
43% answered correctly

Explanation:



When the carbon atoms in two or more alkene functional groups are directly connected to each other, the double bonds are classified as **conjugated**. Importantly, the pi electrons in conjugated alkenes are distributed throughout the conjugated region and are no longer isolated to one alkene functional group. Consequently, the properties and reactivity of conjugated alkenes are different than those for isolated alkenes.

Understanding the properties and reactivity of conjugated alkenes requires application of **molecular orbital (MO) theory**. Since electron orbitals can be described by their wave function (a mathematical expression), the linear combination of atomic orbitals leads to the generation of MOs that distribute electron density throughout the nuclei in a molecule (rather than at one specific atom).

Like atomic orbitals, MOs are populated by electrons starting with the lowest-energy MO. The **highest occupied molecular orbital (HOMO)** is the highest-energy MO containing at least one electron, and the **lowest unoccupied molecular orbital (LUMO)** is the lowest-energy vacant MO.

Delocalization of electrons through conjugation decreases the average energy of filled orbitals. Consequently, **conjugated alkenes** are more stable, of lower energy, and have a **smaller energy difference between** the **HOMO and LUMO** than isolated alkenes.

This question asks what can be inferred about the relative HOMO-LUMO energy difference for α-guaiene and guaiazulene. Guaiazulene is an **aromatic compound** containing a core of alternating single-double bonds—a type of conjugated system. In contrast, α-guaiene contains two isolated (nonconjugated) alkene groups. Because conjugation decreases the HOMO-LUMO energy difference, it can be concluded that **the HOMO-LUMO energy difference is smaller for guaiazulene**.

(Choice B) Because conjugated alkenes are more stable and lower in energy than isolated alkenes, conjugation *decreases* the HOMO-LUMO energy difference. Guaiazulene is conjugated and has a *smaller* HOMO-LUMO energy difference.

(Choices C and D) Although their carbon skeletons are similar, guaiazulene contains an aromatic (conjugated) core whereas α-guaiene does not. As a result, the HOMO-LUMO energy difference for these molecules is *different*, and inferences can be made about the relative magnitude of the HOMO-LUMO gap.

Educational objective:
Conjugated alkenes are comprised of alternating single-double bonds and are more stable than isolated (nonconjugated) alkenes. Application of molecular orbital theory demonstrates that conjugated alkenes have a smaller HOMO-LUMO gap.

Time spent:0 sec **QID**:404133

Subject:
Organic Chemistry

System:
Separation Techniques, Spectroscopy, and Analytical Methods

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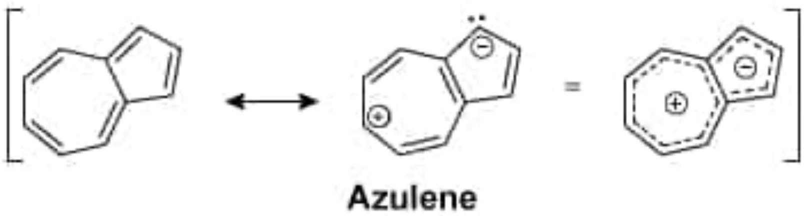

Figure 1 Azulene resonance form

The azulene-related compounds α-guaiene and guaiazulene (Figure 2) are natural products that are biosynthetically derived from the same 15-carbon metabolite, farnesyl pyrophosphate. These two materials are prepared through divergent biosynthetic pathways and have quite different properties. α-Guaiene is one of several nonaromatic, sesquiterpene isomers isolated from guaiac wood oil that have found use within the flavoring and fragrance industries.

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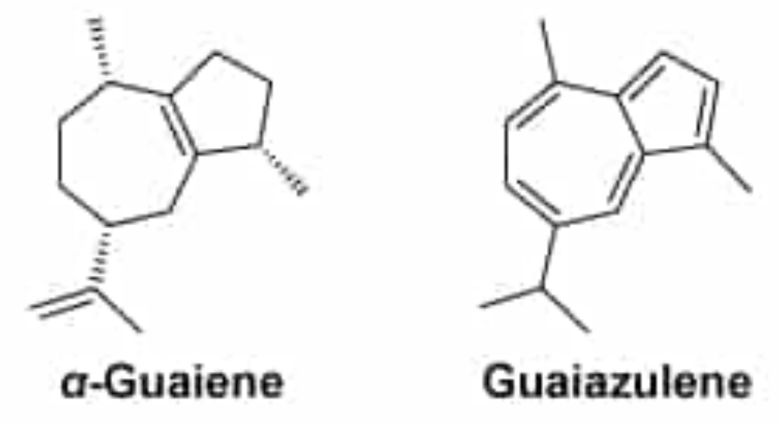
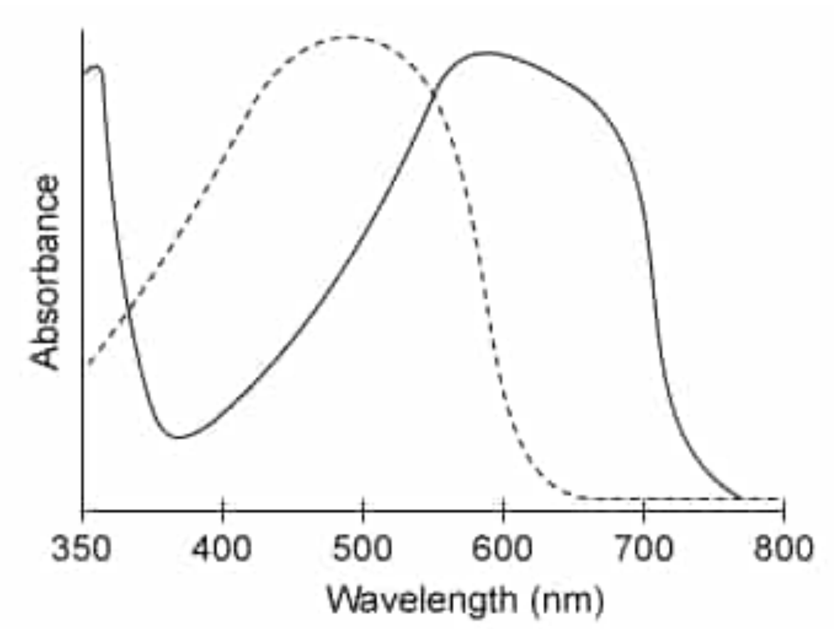


Figure 2 Azulene-related natural products

The following figure shows the visible spectrum for azulene (solid line) and an azulene derivative (dotted line).

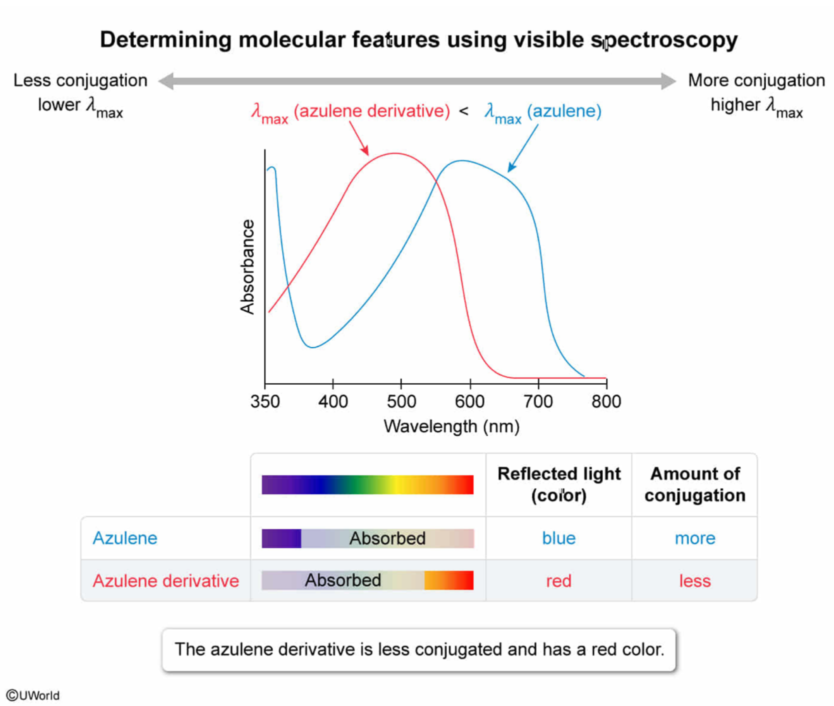


Based on this information, which of the following statements most accurately describes the qualities of the azulene derivative?

- ◯ A. The azulene derivative is more conjugated than azulene and has a blue color. [21%]
- ✔ ◉ B. The azulene derivative is less conjugated than azulene and has a red color. [35%]
- ◯ C. The azulene derivative is more conjugated than azulene and has a red color. [17%]
- ◯ D. The azulene derivative is less conjugated than azulene and has a blue color. [25%]

Correct
35% answered correctly

Explanation:



Understanding the properties of conjugated molecules requires application of **molecular orbital (MO) theory**. Like atomic orbitals, MOs are populated by electrons starting with the lowest-energy MO. Consequently, conjugated functional groups are more stable and of lower energy than isolated functional groups.

Photons in the ultraviolet (UV) and visible regions of the electromagnetic spectrum have the correct energy for absorption by conjugated molecules. Photon absorption promotes an electron in the ground state (highest occupied molecular orbital, or **HOMO**) to an excited state (lowest unoccupied molecular orbital, or **LUMO**). The energy of a photon is proportional to its frequency and inversely proportional to its wavelength. Since conjugation decreases the size of the HOMO-LUMO energy gap, **conjugation decreases** the **energy** of **absorbed photons** and **increases the wavelength of maximum absorption λ_{max}**.

As the **UV-visible spectrum** of highly conjugated molecules becomes shifted to increasingly higher wavelengths, portions of the visible light spectrum are absorbed. The wavelengths (colors) of light *not* absorbed are reflected back to the viewer and become the apparent color of a substance. Therefore, absorption and reflection spectra for a substance are reciprocal to each other.

Compared to azulene (solid line), the visible spectrum of the **azulene derivative** (dotted line) provided in this question is **shifted to lower wavelengths**. According to MO theory, a decrease in the λ_{max} corresponds to a **decrease in conjugation** versus azulene. Furthermore, the azulene derivative λ_{max} (460 nm, violet/blue light) and absorption minima (higher than 570 nm, orange/red light) suggest that the azulene derivative has a **red color**.

Therefore, the **azulene derivative** must be **less conjugated** than azulene, and has a **red color**.

(Choice A) Shifting a molecule's visible spectrum to lower wavelengths is associated with both *less conjugation* and an increase in apparent *red* color.

(Choice C) A visible spectrum shift to lower wavelengths indicates that the size of the HOMO-LUMO energy gap is *larger* for the azulene derivative. Consequently, the azulene derivative must be *less* conjugated than azulene.

(Choice D) The azulene derivative has an absorption *maximum* in the visible region of blue light (450–490 nm). Blue photons are *absorbed* rather than reflected as part of the molecule's apparent color.

Educational objective:
Conjugation decreases the size of the HOMO-LUMO energy gap and leads to the absorption of photons with a lower energy. The visible spectrum can be analyzed to gain information about the nature of conjugation and the molecule's apparent color.

Time spent:0 sec

Subject:
Organic Chemistry

QID:404134

System:
Separation Techniques, Spectroscopy, and Analytical Methods

Azulene is an organic natural product comprised of fused seven- and five-membered rings containing a conjugated pi system. The aromaticity of azulene's 10π electron system is most evident when examining its resonance form, which is represented by fused aromatic tropylium and cyclopentadiene ions (Figure 1). This electronic feature is primarily responsible for azulene's abnormally large dipole moment for a hydrocarbon (1.08 D), differential chemical reactivity of its two rings, and striking, deep blue color. As a naphthalene isostere, azulene-containing compounds have been identified with anti-inflammatory and anti-infectious (bacteria, fungal, viral, cancer) properties. The unique molecular and electronic scaffold of azulene continues to make it an interesting target within the drug development and optoelectronic communities.

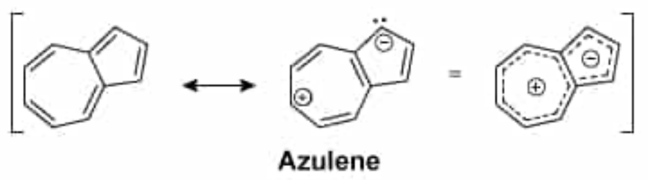


Figure 1 Azulene resonance form

The azulene-related compounds α-guaiene and guaiazulene (Figure 2) are natural products that are biosynthetically derived from the same 15-carbon metabolite, farnesyl pyrophosphate. These two materials are prepared through divergent biosynthetic pathways and have quite different properties. α-Guaiene is one of several nonaromatic, sesquiterpene isomers isolated from guaiac wood oil that have found use within the flavoring and fragrance industries. In contrast, guaiazulene is an alkylated azulene variant that shares a common aromatic core and electronic properties. Guaiazulene-derived compounds have been developed in recent years as potential treatments for a wide range of diseases and conditions.

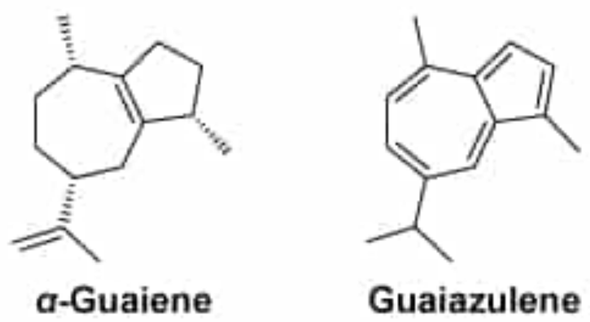


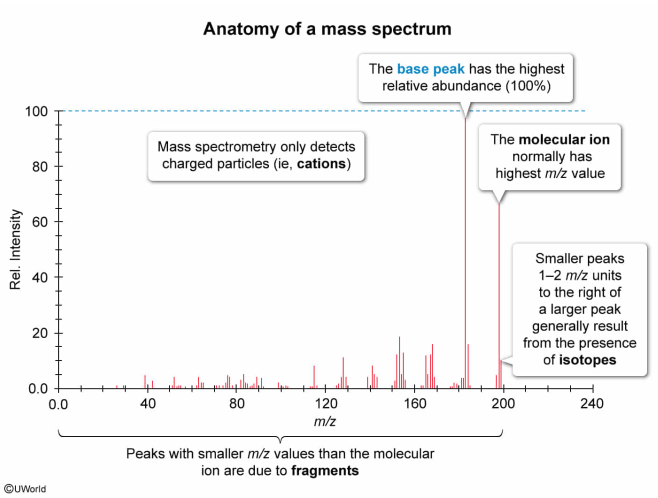
Figure 2 Azulene-related natural products

In mass spectrometry, which of the following statements most accurately describes the definition of the base peak?

- ✔ A. The peak with the highest abundance [45%]
- B. The peak with the highest *m/z* value [29%]
- C. The peak with the lowest abundance [6%]
- D. The peak with the lowest *m/z* value [18%]

Correct
46% answered correctly

Explanation:



An electron-impact mass spectrometer bombards a sample with electrons to remove an electron from a molecule (**ionization**). This creates the **molecular ion**, a radical cation with a **mass-to-charge ratio (*m/z*)** equal to the molecular weight of the molecule. Normally, the peak with the highest *m/z* value corresponds to the molecular ion.

The molecular ion can also break into a radical and a cation to produce smaller molecular fragments. Ionized material enters a magnetic field, which curves the path of the particles by their *m/z* value. Only **cations** with *m/z* values that match the curvature of the instrument tube reach the detector. Finally, a **mass spectrum** is generated.

The abundance of cations detected at each *m/z* value is variable and depends on the frequency (likelihood) of fragmentation patterns in a population, the stability of cations produced, or a combination of factors. The *y*-axis in a mass spectrum (abundance) is reported on an arbitrary 0 to 100% scale, where the highest abundance peak is given a relative abundance of 100%. The highest abundance peak (called the **base peak**) corresponds to the cation with the greatest stability and/or relative rate of formation, and it can provide structural clues toward the identity of the original molecule.

Therefore, the statement that most accurately describes the definition of base peak in mass spectrometry is **the peak with the highest abundance**.

(Choice B) The peak with the highest *m/z* value is normally the *molecular ion*, which is the original molecule less the mass of one electron.

(Choices C and D) The use of "base" in the phrase *base peak* is inaccurate, as it does not reflect low values found on either axis of a mass spectrum. Neither of these choices are particularly useful in mass spectrometry: Low abundance peaks become difficult to differentiate from background noise produced by the instrument detector whereas the lowest *m/z* peak (ie, *m/z* 1) corresponds to a hydrogen cation (proton) that would be generated by most organic molecules.

Educational objective:
Mass spectrometry experiments ionize samples and detect the abundance of ions at each mass-to-charge ratio (*m*/*z*). The base peak is the *m/z* value that produces the highest relative abundance (100%) and normally corresponds to a cation that is either especially stable or likely to form.

Time spent:0 sec

Subject:
Organic Chemistry

QID:404135

System:
Separation Techniques, Spectroscopy, and Analytical Methods

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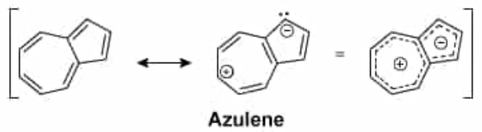

Figure 1 Azulene resonance form

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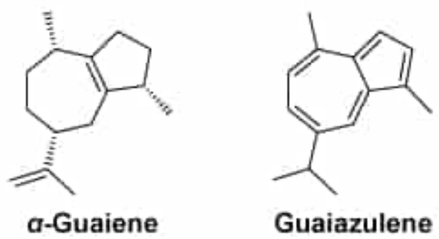
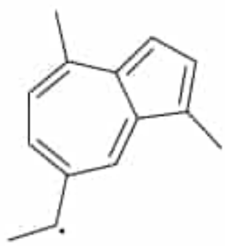


Figure 2 Azulene-related natural products

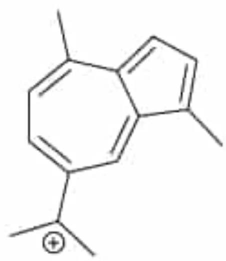
If the molecular weight of guaiazulene is 198 g mol^{-1}, which of the following structures represents a contributing ion to the *m/z* 183 peak in its mass spectrum?

○ A.



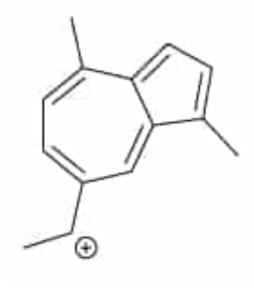
[14%]

○ B.



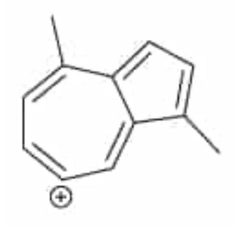
[17%]

C.



[60%]

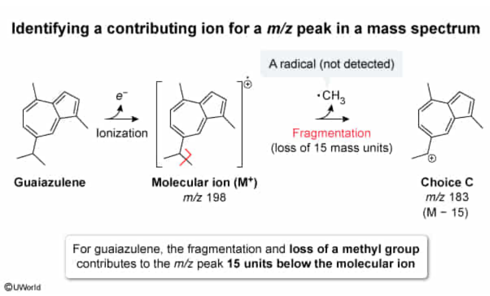
D.



[7%]

Correct
61% answered correctly

Explanation:



Mass spectrometry provides the ability to measure the mass of molecules and molecular fragments. An electron-impact mass spectrometer bombards a sample with electrons to remove an electron from a molecule (**ionization**) and create the **molecular ion (M^+)**, a radical cation with a **mass-to-charge ratio (*m*/*z*)** equal to the molecular weight of the molecule. The M^+ can also break into a radical and a cation to produce smaller molecular fragments. The M^+ and fragments are detected and displayed on a **mass spectrum**.

Since mass spectrometry only provides a count of detected ions at each *m/z* value, additional inference is required to correlate mass spectral data to molecular structure. A single *m/z* peak is typically generated

from multiple unique ions with the same *m/z* value. Peaks are described by their **mass difference** from the molecular ion peak (M), where the magnitude of the *m/z* difference can indicate the atom(s) lost during fragmentation. Common mass differences in spectra result from alkyl fragmentation. If the structure of the original molecule is known, the mass difference between the molecular ion and a given *m/z* peak can be correlated to the fragmentation of specific sigma bonds.

In this question, the indicated peak (*m/z* 183) is 15 *m/z* units lower than the molecular ion of guaiazulene (*m/z* 198). An [M − 15] peak is usually attributed to the **loss of a methyl group**. Consequently, **cation** structures that depict a loss of a methyl group from guaiazulene would be detected within the *m/z* 183 peak.

The structure shown in Choice C depicts a cationic fragment of guaiazulene that has lost one of its two methyl groups within the isopropyl group.

(Choice A) Although this structure correctly indicates a fragment that has lost a methyl group, it is a *radical* molecule. Only *cations* are detected by a mass spectrometer.

(Choice B) This tertiary cation can only be generated if the molecular ion fragments and loses a *hydrogen radical* from its isopropyl group. The resulting [M − 1] peak would appear at *m/z* 197, *not m/z* 183.

(Choice D) Generating this cation requires the loss of the *whole* isopropyl group (C_3H_7). The resulting [M − 43] peak would appear at *m/z* 154, *not m/z* 183.

Educational objective:
Mass spectrometry experiments ionize samples and detect the abundance of ions at each mass-to-charge ratio (*m*/*z*). The mass difference between fragment peaks and the molecular ion peak can be correlated to specific types of fragmentation in the molecular structure.

Time spent:0 sec	**QID**:404136
Subject:	**System:**
Organic Chemistry	Separation Techniques, Spectroscopy, and Analytical Methods